University of Trento
Master's Degree in Computer Science

Data Visualization
Technical Report

Data Visualization Lab
A.Y. 2025/2026

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July 2026

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1 City Bike Dataset

Our main source of data is the City Bike dataset, which contains information about bike trips in New York City. These datasets are published on a monthly basis and can be accessed through the Citi Bike website.[1]

The process involved in writing this report can be found in the Jupyter notebook in our repository.[2]

1.1 Dataset Structure

The data is organized in multiple CSV files, each containing information about bike trips that ended in that month. Each row in the dataset represents a single bike trip and includes various attributes that describe the trip.

1.1.1 Original Structure

The original dataset includes the following columns:

- **Ride ID**: A unique identifier for each bike trip.
- **Rideable Type**: The type of bike used for the trip. Can be either "classic bike" or "electric bike".
- **Started At**: a timestamp indicating when the trip started.
- **Ended At**: a timestamp indicating when the trip ended.
- **Start Info**: Information about the station and location where the trip started, structured as:
  - **Station Name**: The name of the starting station.
  - **Station ID**: The identifier of the starting station.
  - **Latitude**: The latitude coordinate of the starting station.
  - **Longitude**: The longitude coordinate of the starting station.
- **End Info**: Information about the destination station, structured similarly to Start Info.
- **User Type**: The type of user: "member" or "casual".

1.1.2 Extracted Features

Additional features were extracted to support the analysis:

- **Trip Duration**: The duration of the trip in seconds, calculated as Ended At − Started At.
- **Trip Distance**: The distance between the start and end stations, computed using the Haversine formula and multiplied by a circuity factor of 1.3 to better approximate real routes.
- **Day of the Week**: The weekday extracted from the start timestamp.

[1] `https://www.citibikenyc.com/system-data`

[2] `https://github.com/davide-dona/nyc-bike-data-visualization/blob/main/src/notebooks/analysis.ipynb`

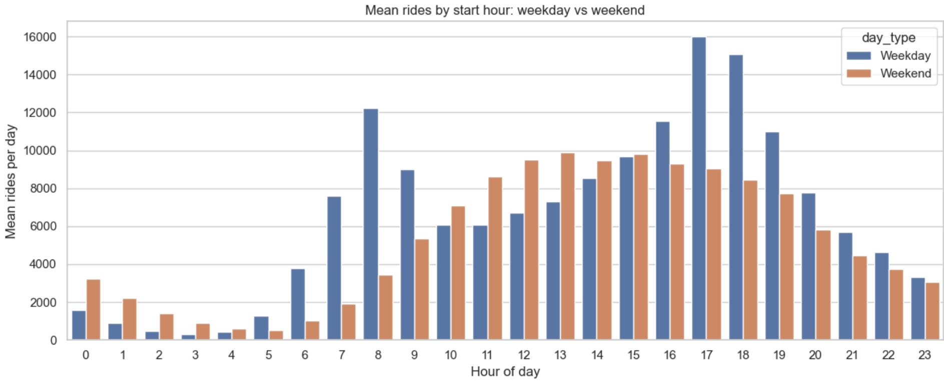

Figure 1: Distribution of trips by hour of the day and day of the week.

- **Hour of the Day**: The starting hour of the trip.
- **Station Popularity**: The number of trips associated with a station within a given time period.
- **Trips Flow**: The number of trips between pairs of stations during a given time period.

1.2 Live Data

Although the City Bike dataset mainly contains historical trips, the analysis is complemented by live station information obtained from the GBFS feed. The live data provides current bike and dock availability, station status, and operational information.

The combination of historical and live data allows the visualization to represent both long-term mobility patterns and the current state of the bike-sharing system.

1.3 Analysis and Insights

The exploratory analysis focuses on temporal, spatial, and behavioural patterns in bike usage.

Temporal analysis shows clear variations in demand depending on the hour of the day and the day of the week. Trips are concentrated during commuting hours, while weekends show different usage patterns associated with leisure activities. Figure 1 summarizes the temporal distribution of bike trips.

Spatial analysis highlights the distribution of trips across the city. Station popularity and trip flows reveal that demand is concentrated around Manhattan and other high-density areas, with some stations acting as important origins and destinations. The spatial distribution of station activity is shown in Figure 2.

User behaviour analysis compares members and casual users. Members generally show more regular commuting patterns, while casual users tend to have shorter-term and more variable usage patterns, as shown in Figure 3.

Trip duration and distance distributions provide additional insight into mobility behaviour. Most trips are relatively short, while longer rides are less frequent and often correspond to recreational usage. The distributions of trip duration and distance are shown in Figures 4 and 5.

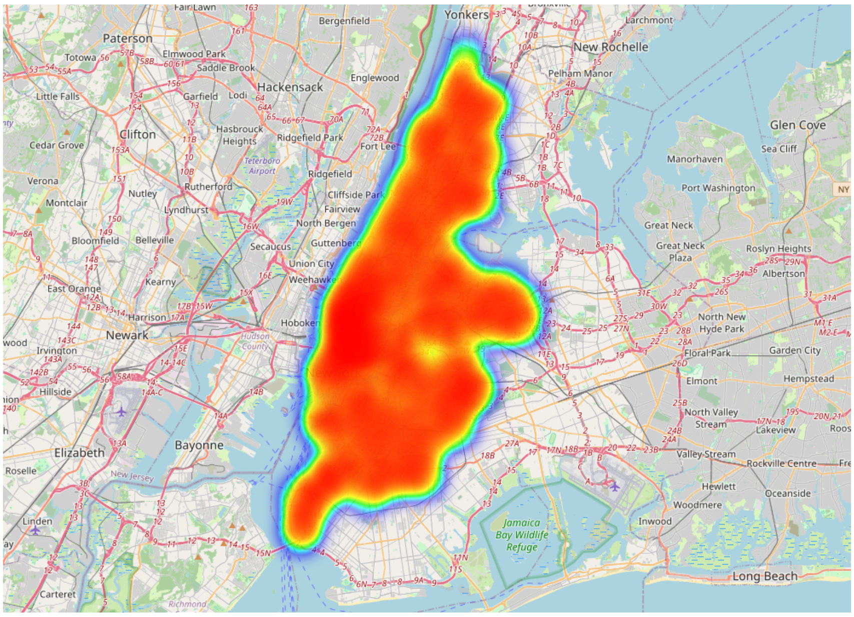

Figure 2: Distribution of trips by station.

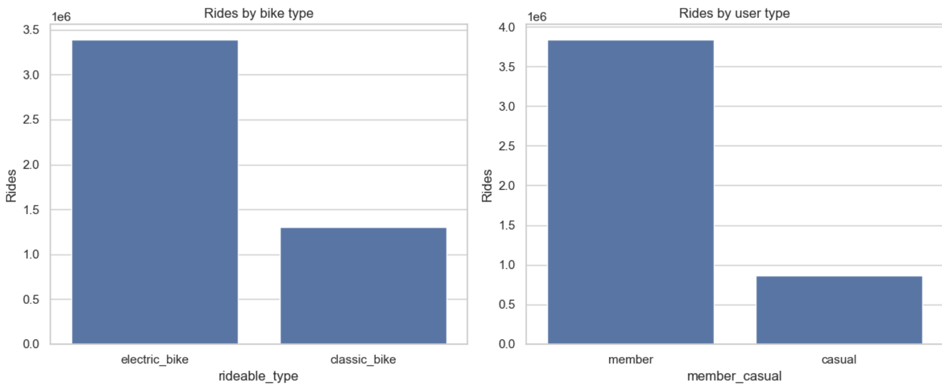

Figure 3: Distribution of trips by user type.

These analyses motivate the use of temporal aggregation, station-level summaries, and flow visualizations in the final dashboard.

1.4 Data Quality

Various data quality issues were encountered during the analysis of the City Bike dataset.

Different Structure During the analysis, we noticed that older datasets have a different structure compared to newer ones. Older datasets may miss columns introduced in newer versions, such as "Rideable Type" or station-related fields, while including deprecated attributes such as Gender, Birth Year, and Bike ID.

This inconsistency required adjustments in the preprocessing pipeline to ensure that all monthly files produced compatible features.

Missing Values and Inconsistencies Several data quality issues were considered during preprocessing.

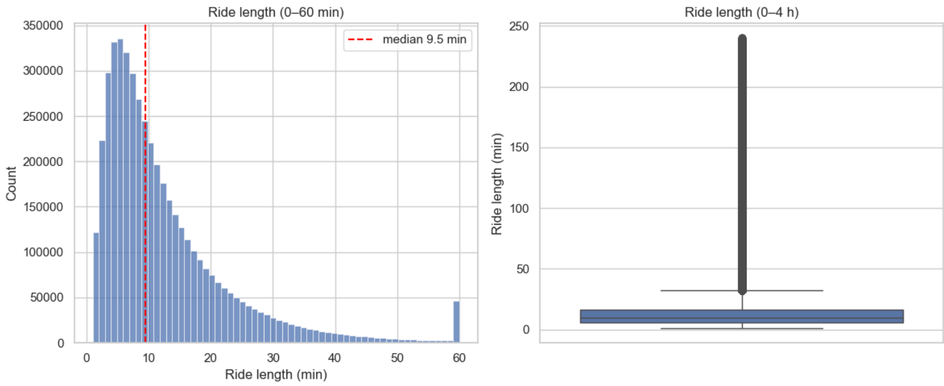

Figure 4: Distribution of trip durations.

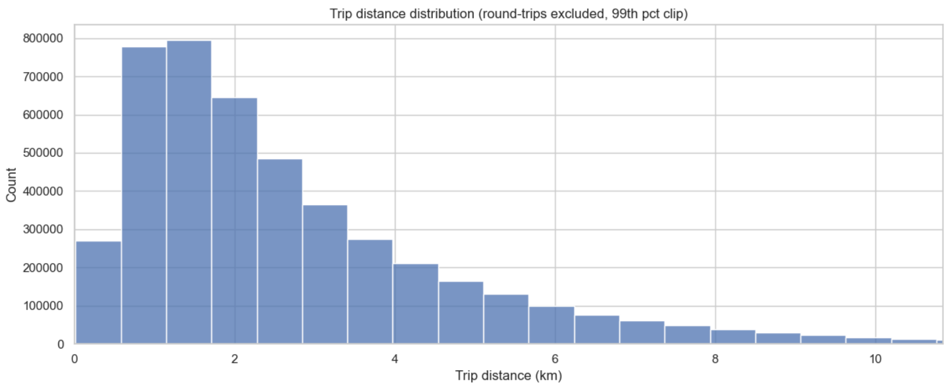

Figure 5: Distribution of trip distances.

Some trips contain unrealistic durations, usually caused by incorrect timestamps, and were removed using reasonable duration thresholds. Trips starting and ending at the same station were also filtered when required, as they can distort distance-based analyses.

A limitation of the dataset is that it only represents Citi Bike users and therefore may not fully describe all cycling activity in New York City. Additionally, trip distance is estimated from station coordinates using the Haversine formula rather than the actual travelled path, making it an approximation.

Station identifiers also require attention. The datasets contain both a numeric station ID and a public short name, but these identifiers are not always consistent across sources and years. Therefore, station joins are performed using the short name after normalization.

Finally, live station data may contain inactive stations or temporary inconsistencies in bike availability. These records are filtered using operational flags before being used in visualizations.

2 Weather Dataset

Our second data source is hourly NYC weather, retrieved from the Open-Meteo archive API.[3]

[3] `https://open-meteo.com/`

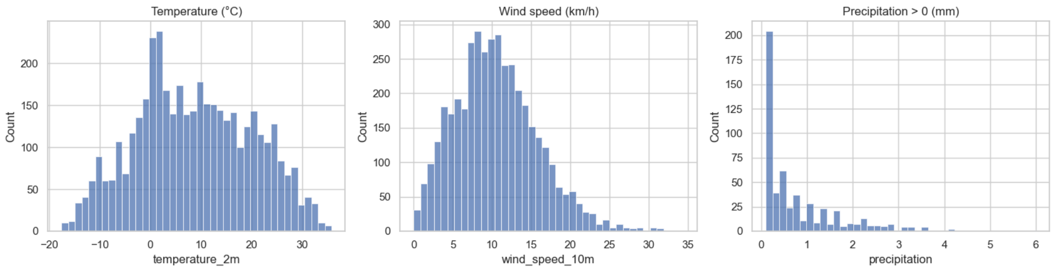

Figure 6: Temperature, wind speed and precipitation distributions over time.

2.1 Dataset Structure

The dataset contains hourly weather observations for New York City, aligned to the `America/New_York` timezone. It provides environmental variables used to explain changes in cycling demand.

2.1.1 Original Structure

The dataset includes the following main fields:

- **Timestamp**: hourly observation time.
- **Temperature (temperature_2m)**: air temperature in °C.
- **Precipitation**: hourly precipitation in millimetres.
- **Weather Code**: WMO weather condition code.
- **Wind Speed (wind_speed_10m)**: wind speed in km/h.
- **Year**: derived field used to partition the dataset into annual Parquet files.

The dataset is used almost entirely in its original form, with only the `Year` column added during preprocessing. Weather data are retrieved from the Open-Meteo archive for the same period covered by the trip dataset.

2.2 Analysis and Insights

The exploratory analysis focuses on weather distributions and their relationship with ridership.

Temperature follows a clear annual cycle, while precipitation is highly right-skewed: most hours record no rainfall, with occasional heavy events. Consequently, precipitation histograms focus on non-zero observations. Wind speed shows moderate variation throughout the year. Figure 6 summarizes the main weather distributions and their seasonal behaviour.

Hourly trip counts are joined with weather observations using their timestamps. The resulting analysis shows that ridership, as expected, is higher during warmer periods and lower during cold or inclement weather. Two plateaus are observed between 15 °C and 20 °C, and between 25 °C and 30 °C. With higher temperatures, ridership increases noticeably, as shown in Figure 7.

These results justify including weather as an explanatory variable for temporal variations in bike usage.

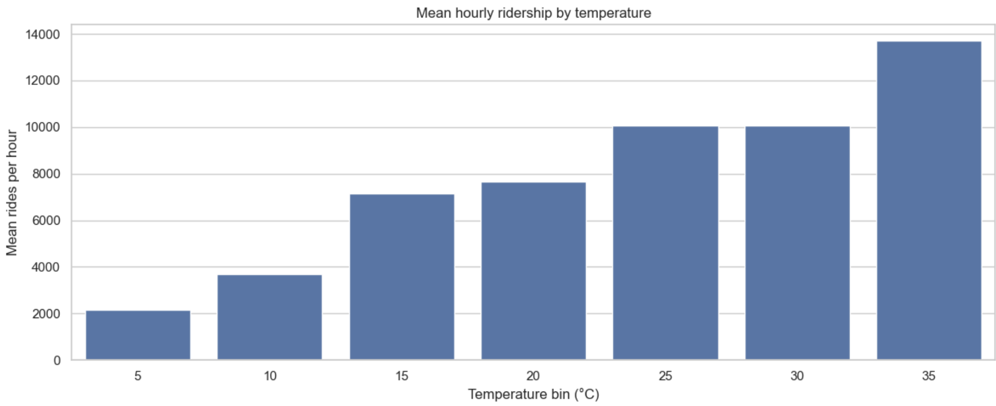

Figure 7: Ridership vs. temperature.

2.3 Data Quality

The archived weather data are complete and consistently sampled at hourly intervals, with no significant missing values.

The main limitation is spatial coverage: the dataset represents weather at a single observation location rather than capturing microclimate differences across the five boroughs. While this may introduce some bias, it is a common approximation for city-scale analyses.

Precipitation is strongly skewed, so non-zero values are analysed separately from dry hours. The derived `Year` column is used only for data partitioning, while additional features such as temperature bins are created only when required for specific analyses.

3 Bike Lanes Dataset

Our third data source is NYC's bike-route network, published by NYC OpenData[4] as dataset `mzxg-pwib`. It provides the city's cycling infrastructure and forms the infrastructure layer of the final visualization.

3.1 Dataset Structure

The dataset describes New York City's bicycle infrastructure at the segment level. Each row represents a physical bike-lane segment, including its geometry, location, status, installation history, and facility type.

3.1.1 Original Structure

The original CSV contains one row per route segment and includes the following relevant attributes:

- **Segment ID**: unique identifier of the route segment.
- **Bike ID**: identifier used to track the segment across revisions.
- **Status**: whether the segment is active or retired.
- **Installation Date**: date the segment entered service.

[4] `https://data.cityofnewyork.us/d/mzxg-pwib`

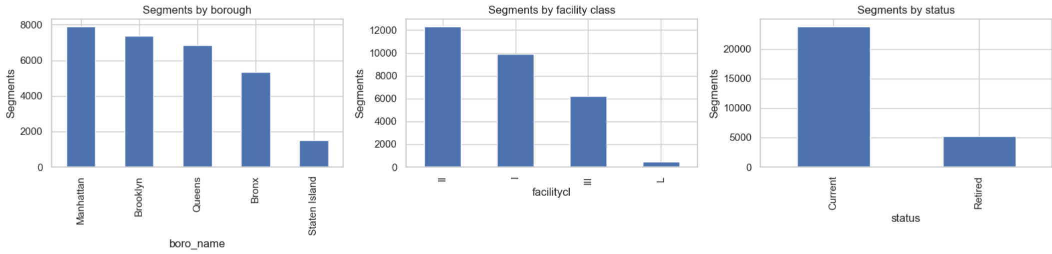

Figure 8: Breakdown of bike-lane segments by borough, facility class and status.

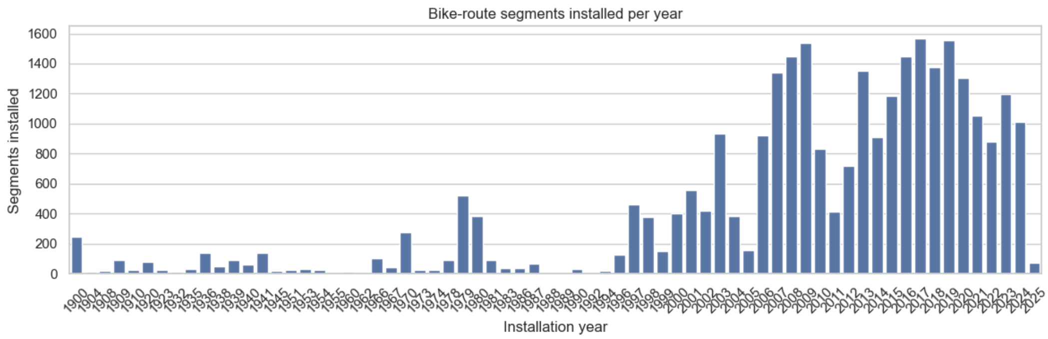

Figure 9: Bike-lane installation timeline.

- **Retirement Date**: date the segment was removed, if applicable.
- **the_geom**: line geometry used for map rendering.
- **Street**, **From Street**, **To Street**: street location.
- **Facility Class**: type of cycling infrastructure.
- **Borough**: borough containing the segment.

The analysis derives summaries by borough, status, and facility class. Unused fields (`prevbikeid`, `gwsys2`, `spur`, `ft2facilit`, and `tf2facilit`) were removed during preprocessing.

3.2 Analysis and Insights

The analysis focuses on four aspects that directly support the final visualization.

A breakdown by borough shows that bike-lane segments are concentrated in Manhattan and Brooklyn, while the remaining boroughs contain fewer segments. A second breakdown compares segment status, showing that most infrastructure is currently active. Figure 8 summarizes the distribution of segments by borough, facility class, and status.

The facility-class distribution distinguishes protected lanes from shared facilities and is used to style the network in the visualization.

The installation timeline illustrates the gradual expansion of the bike network over time, reflecting continued investment in cycling infrastructure. Figure 9 shows the number of installed segments over the years.

3.3 Data Quality

The dataset is generally clean but requires limited preprocessing.

Unused columns were removed to simplify the analysis without affecting the visualization.

Current and retired segments are treated separately so that obsolete infrastructure is not included in the active network layer.

Installation and retirement dates are occasionally missing, causing slight underestimation in historical summaries.

Records with missing or invalid geometries cannot be displayed and are filtered before mapping. For the interactive visualization, only a sample of segments is rendered to improve responsiveness while preserving the overall spatial pattern.

4 Station Metadata Dataset (GBFS)

Our fourth source of data is the Lyft General Bikeshare Feed Specification (GBFS) feed,[5] which provides current station information and live availability. Two endpoints are merged: `station_information` (static metadata) and `station_status` (real-time availability).

4.1 Dataset Structure

The station metadata dataset combines static station properties with live operational information. The static endpoint describes location and capacity, while the live endpoint reports availability and station status.

4.1.1 Original Structure

The merged dataset contains:

- **From `station_information` (static):**
  - **Station ID**: numeric GBFS identifier.
  - **Short Name**: public station identifier used for joins with Citi Bike trip data.
  - **Name**: station name.
  - **Latitude** and **Longitude**: station coordinates.
  - **Capacity**: number of available docks.
- **From `station_status` (live):**
  - **Bikes available**: currently rentable bikes.
  - **Docks available**: currently available docks.
  - **Operational flags**: `is_installed`, `is_renting`, and `is_returning`.

4.1.2 Extracted Features

An `active` indicator is derived from the three operational flags and is used to filter unavailable stations from live views.

[5] `https://gbfs.lyft.com/gbfs/2.3/bkn/en/gbfs.json`

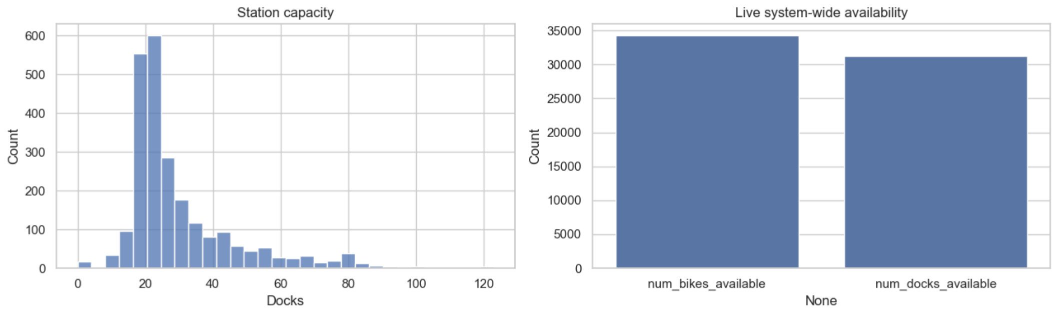

Figure 10: Distribution of station capacity (number of docks) and live availability (bikes and docks available).

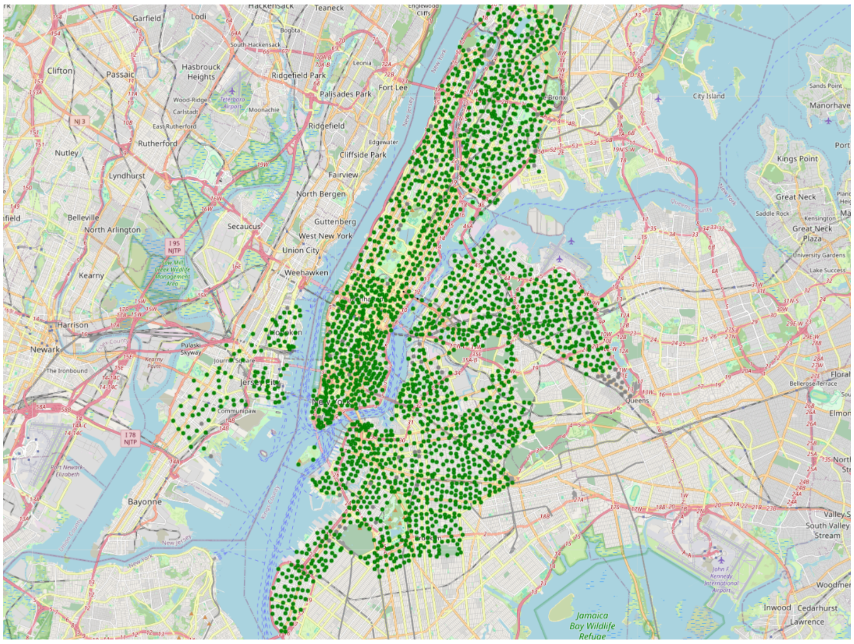

Figure 11: Map of stations in New York City, colored by their active state.

4.2 Analysis and Insights

The analysis focuses on station capacity, live availability, and spatial distribution.

Capacity distributions show how the network is provisioned, distinguishing small stations from larger hubs. Live availability of bikes and docks highlights possible imbalances, such as empty or saturated stations. The corresponding distributions are shown in Figure 10.

The station map provides a spatial overview of coverage and density, helping identify clusters, gaps, and possible coordinate issues. Capacity and availability values can also be used as visual encodings to improve the final dashboard, as shown in Figure 11.

4.3 Data Quality

The main quality issues concern station status, missing attributes, and identifier consistency.

Some stations are present in the feed but not fully operational because one or more status flags are false. These stations are excluded from live visualizations using the derived `active` feature.

A small number of records may have missing capacity or coordinates. Stations without coordinates cannot be displayed on maps, while missing capacity only affects analyses based on station size.

Cross-dataset joins require attention because GBFS provides both a numeric `station_id` and a public `short_name`. Historical trip data use `short_name`, therefore all joins are performed using this field after validating its consistency.

Additional issues, such as temporary feed outages or duplicate identifiers caused by station changes, are handled through preprocessing checks to maintain reliable visualizations.